

Federated Learning for Network Intrusion Detection: A Survey of Backdoor Attacks and Defenses

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Abstract—Federated Learning (FL) has emerged as a promising privacy-preserving paradigm for training Network Intrusion Detection Systems (NIDS) by enabling collaborative model development without centralizing sensitive network traffic data. However, the distributed architecture of FL introduces critical vulnerabilities to backdoor attacks, where malicious clients can implant hidden triggers that cause the global model to misclassify attacker-controlled traffic while maintaining normal performance on benign data. This survey provides a comprehensive analysis of backdoor attacks and defenses in federated learning, with a specific focus on their applicability to network intrusion detection. We categorize the evolving landscape of backdoor threats from simple data poisoning to sophisticated parameter poisoning and adaptive attacks and examine state-of-the-art defense mechanisms, including anomaly detection, robust aggregation, and differential privacy techniques. Through critical analysis, we highlight major research limitations, including the dependence on unrealistic image-based benchmarks, the weakness of anomaly-based defenses under non-IID conditions, and the lack of backdoor-focused evaluation metrics. In general, our findings indicate that, despite significant theoretical advances, current defenses remain largely unvalidated for real-world NIDS deployments. This survey provides a foundation for future research aimed at developing practically robust, privacy-preserving federated NIDS that can withstand sophisticated backdoor attacks in heterogeneous network environments.

Index Terms—Federated Learning, Network Intrusion Detection System (NIDS), Backdoor Attacks, Poisoning Attacks, Cybersecurity, non-IID data

I. INTRODUCTION

In the modern digital landscape, the security of computer networks is paramount. Network Intrusion Detection Systems (NIDS) serve as the first line of defense, monitoring network traffic to identify and mitigate cyber threats. Traditionally, the most effective NIDS are built using machine learning (ML) models trained on vast and diverse datasets of network traffic [1]. However, this centralized approach presents a fundamental conflict: to be effective, the NIDS must analyze all network data, yet this very centralization creates a massive privacy and security liability, requiring organizations to share sensitive, proprietary or private user data with a central server.

Federated Learning (FL) has emerged as a transformative solution to this privacy paradox. FL is a distributed ma-

chine learning paradigm that enables multiple clients such as different organizations, branches, or even IoT gateways to collaboratively train a robust global model without ever sharing their raw local data [2]. Instead, each client trains a model locally and only the resulting model parameters (or "updates") are sent to a central server for aggregation. This approach has shown immense promise for privacy-preserving NIDS, allowing for the creation of sophisticated threat detection models that leverage data from many networks while preserving the confidentiality of each participant [3], [4].

However, this distributed architecture introduces a new and critical attack vector. By giving clients control over their local training, FL inherently trusts them to provide honest model updates. Malicious clients can exploit this trust to launch poisoning attacks, intentionally submitting corrupted updates to degrade or control the global model [5]. Among these, backdoor attacks represent the most insidious and stealthy threat [6]. A backdoor attack is a targeted form of poisoning where the global model's performance on normal tasks remains high, but the model learns to misbehave in a specific way chosen by the attacker when it encounters a secret "trigger." In an NIDS, this could allow an attacker to create a "master key," permitting their own malicious traffic to pass undetected.

This threat has given rise to a critical field of research focused on building defenses against backdoor attacks in FL. These defenses generally fall into three main categories: (1) robust aggregation algorithms that statistically minimize the impact of malicious updates, (2) anomaly detection methods that inspect client updates and filter out outliers, and (3) differential privacy techniques that add noise to obscure malicious contributions [6].

Despite rapid advancements in federated learning security, a significant gap remains between the theoretical development of backdoor defenses and their practical deployment in real-world NIDS environments. Much of the prior work such as FedDefender [7] and CrowdGuard [8] has been evaluated on image-based datasets like MNIST or CIFAR-10. However, network intrusion detection data differs fundamentally: it is high-dimensional, tabular, sparse, and inherently non-IID due to diverse traffic patterns generated across different parts of

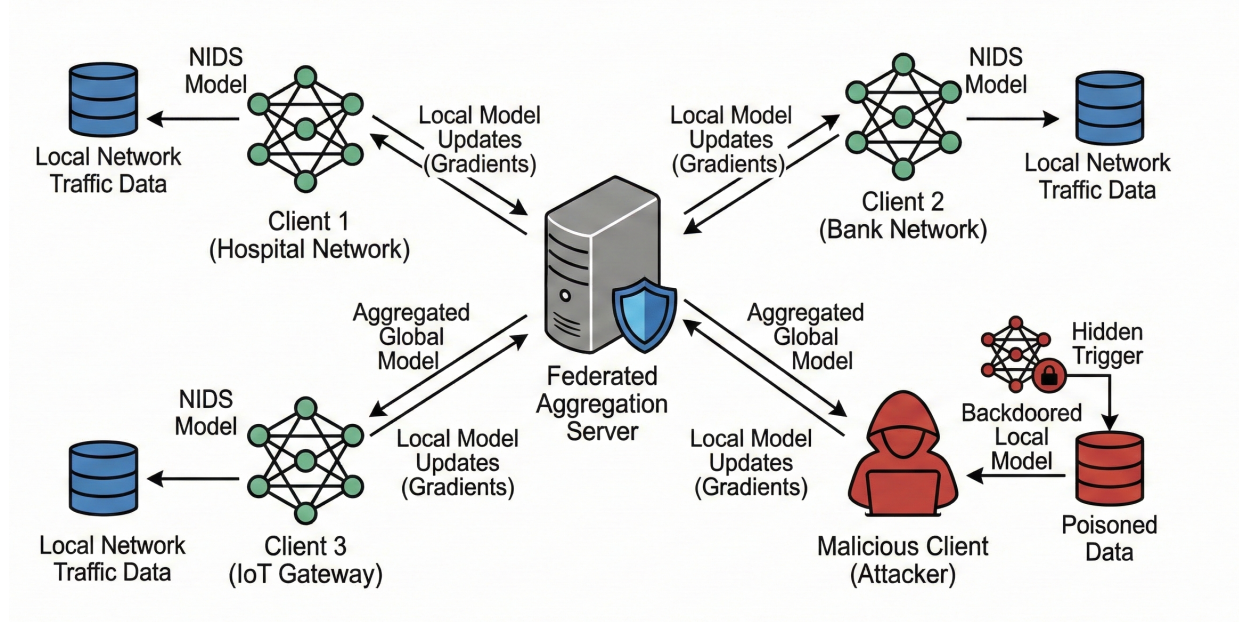


Fig. 1. Overview of a Federated Learning-based NIDS under backdoor attack and defense scenarios

a network. Several efforts have begun addressing security in federated NIDS such as SecFedNIDS [9] for poisoning attacks and FLAME [10] and SHIELD-FL [11] for backdoor resilience. However, the domain still lacks comprehensive, practically validated defenses. Many existing anomaly-based detection strategies that rely on the assumption of similarity among benign clients are unlikely to hold in realistic NIDS deployments.

This literature review aims to systematically categorize and analyze the current landscape of backdoor attacks and defenses in federated learning. We will then critically examine this literature to highlight the existing research gap: the unverified and questionable effectiveness of current defenses when applied to the complex, non-IID data characteristic of network intrusion detection. This analysis will map the "state-of-the-art" and identify the crucial future work required to build a federated NIDS that is not only private, but also genuinely secure.

II. THREAT MODEL: THE EVOLVING LANDSCAPE OF BACKDOOR ATTACKS

To design effective defenses, it is first necessary to understand the adversary. In the context of Federated Learning (FL), an attacker is typically a malicious client (or a collusion of clients) participating in the training process. Their goal is not to simply degrade the global model's accuracy, which would be easy to detect, but to implant a "backdoor." A backdoored model performs normally on benign data, but when it encounters a specific, secret "trigger", a pattern known only to the attacker, it produces an incorrect, attacker-chosen output [6].

This section reviews the literature on the evolution of these attacks, from simple data poisoning to highly adaptive, stealthy assaults.

A. Foundational Poisoning Attacks in FL-NIDS

The most straightforward attack is data poisoning. In the context of an FL-based NIDS, an attacker can launch a label-flipping attack by poisoning their local dataset. For example, by taking samples of a specific "attack" traffic (e.g. a "DDoS" attempt) and re-labeling them as "benign" [12]. When the local model is trained on this poisoned data, it learns to misclassify that specific attack. As demonstrated by [24] and [25], even simple poisoning attacks can significantly degrade the performance of an FL-based NIDS. However, these naive attacks are often "noisy" and create a local model update that deviates significantly from the benign clients, making them susceptible to basic anomaly detection.

B. Stealthy and Targeted Backdoor Attacks

Recognizing this, researchers have developed far more advanced "stealth" attacks designed to be indistinguishable from normal training. The threat has evolved from broad, noisy attacks to surgical, adaptive ones.

Distributed and Client-Targeted Attacks: The threat is not limited to a single trigger. [17] proposed a Distributed Backdoor Attack (DBA) where the global trigger is decomposed into localized sub-triggers, making it exceptionally difficult to detect from any single client's update. Going further, [22] introduced the "Sniper Backdoor," a highly personalized attack that targets only a single, specific client in the FL network, leaving the global model and all other clients untouched. This evades any defense that relies on detecting colluding or anomalous groups of clients.

Adaptive Attacks: Modern attackers do not use a fixed, static trigger. [19] developed A3FL, an "Adversarially Adaptive Backdoor Attack" where the attacker dynamically optimizes the backdoor trigger itself. The trigger is designed to

TABLE I
COMPREHENSIVE TAXONOMY OF BACKDOOR AND POISONING ATTACKS IN FEDERATED NIDS

Attack Type	Sub-Category	Attack Name	Mechanism	Scope	Stealth	Ref
Data Poisoning	Label Manipulation	Label Flipping	Flips labels (e.g., Attack $\rightarrow$ Benign) to degrade accuracy.	Global	Low	[12]
	Trigger Injection	Naive Backdoor	Stamps a fixed pixel/feature pattern on inputs.	Global	Med	[13]
		PCAP-Backdoor	Injects trigger packets into raw network traffic (PCAP).	Global	High	[14]
		Feature-Imp.	Modifies packet features based on importance scores.	Global	High	[15]
Model Poisoning	Param. Manip.	Param. Poisoning	Modifies critical weights (Key Parameters) directly.	Global	V. High	[16]
	Trigger Decomp.	DBA	Decomposes global trigger into local sub-triggers.	Global	High	[17]
	Collab. Synergy	CollaPois	Coordinates gradients to exploit Non-IID distribution.	Global	High	[18]
	Adaptive Opt.	A3FL	Adversarially optimizes updates to resist unlearning.	Global	V. High	[19]
	Generative (GAN)	VulnerGAN	Uses GANs to generate vulnerability-amplifying gradients.	Global	High	[20]
		PI-SBA	Uses GANs to reconstruct data for stealthy poisoning.	Global	High	[21]
	Targeted		Targeted attack affecting only a specific victim client.	Targeted	V. High	[22]
	Vertical FL (VFL)	BadVFL	Poisons VFL embeddings without access to labels.	Global	High	[23]

be "less likely to be removed by the global training dynamics," effectively making it robust against the learning process itself.

Parameter-Poisoning Attacks: Perhaps the most advanced threat bypasses data poisoning entirely. [16] demonstrated a backdoor attack that works by "Poisoning Key Parameters." Here, the attacker does not need to manipulate their local data; they directly manipulate a small, critical subset of their local model's parameters before sending the update. This allows the attack to be highly effective and nearly invisible, as the overall parameter update can be crafted to look statistically identical to a benign one.

C. Attacks in Different FL Contexts

The threat also extends beyond the standard, horizontal FL setting. [23] demonstrated the feasibility of backdoor attacks in Vertical Federated Learning (VFL), a more complex scenario where clients share the same set of instances but have different features. This attack, dubbed "BadVFL" [23], is particularly challenging as the attacker does not even have access to the data labels, further illustrating the sophistication and adaptability of modern adversaries.

In summary, the threat model is not a simple data-poisoning problem. The modern adversary is adaptive, precise, and can target data, parameters or even specific clients. Any proposed defense for an FL-based NIDS must be robust enough to handle these stealthy and evolving attacks, not just the simpler poisoning attacks of the past.

III. STATE-OF-THE-ART BACKDOOR DEFENSES IN FEDERATED LEARNING

In response to the sophisticated threats outlined in Section 2, the research community has proposed numerous defense

mechanisms. These defenses aim to ensure the integrity of the global model by identifying and mitigating the impact of malicious client updates. The majority of these strategies can be classified into three primary categories: (1) anomaly detection-based defenses, which filter out malicious updates; (2) robust aggregation algorithms, which statistically minimize the impact of malicious updates and (3) privacy-enhancing techniques, which incidentally provide robustness by obscuring individual contributions.

A. Anomaly Detection-Based Defenses

The most intuitive and popular defense strategy is to treat the central server as a "bouncer" that inspects all incoming client updates and rejects those deemed malicious before they can be aggregated. These methods rely on the assumption that malicious updates, even stealthy ones, will exhibit statistical properties that differentiate them from benign updates.

A benchmark for this approach is FLAME [10]. FLAME operates on the principle of clustering, utilizing the cosine similarity of client updates to group them. It assumes that benign clients will form a large, dense cluster, while malicious clients will be statistical outliers. By filtering out these outliers before aggregation, FLAME can effectively tame backdoor attacks.

Many other papers have explored this "outlier detection" concept. For instance, [30] also use cosine similarity to detect and isolate malicious agents. [31] propose BADFL, which combines clustering with interpretability techniques to more accurately identify malicious models.

Beyond simple clustering, researchers have developed defenses targeting specific anomalies in model behavior.

TABLE II
BACKDOOR DEFENSES IN FEDERATED LEARNING

Defense	Phase	Type	Core Mechanism	Privacy	Non-IID	Limitations
SHIELD-FL [11]	Pre-agg.	Anomaly Det.	Detect abnormal neurons via L2-norm changes after clean de-learning; prune before sending updates	Standard	Yes	Needs small clean dataset; fails if majority malicious; pruning may remove useful neurons; limited evaluation
FLAME [10]	In-agg.	Anomaly + DP	Dynamic model filtering (clustering); adaptive weight clipping; DP-based noise	DP-based	Yes	Fails when $PMR \geq 50\%$; weak evaluation under multiple non-IID scenarios
Deepsight [26]	Pre-agg.	Anomaly Det.	Inspect model metrics (DDifs, NE-UPs, thresholds); cluster models; voting-based filtering; dynamic cluster-wise clipping	Standard	Yes	Computation-heavy; needs majority-benign assumption; limited non-IID evaluation
FedDefender [7]	Pre-agg.	Anomaly Det.	Compare neuron activations using random synthetic inputs; adjust/discard abnormal clients via nk	No	No	Aggressive threshold tuning needed; not evaluated under non-IID; assumes single malicious client; activation mimicry can bypass
FLEDGE [27]	In-agg.	Hybrid	CKKS HE + blockchain to secure updates; G-KDE detects poisoned updates; reward-penalty smart contracts	Standard	No	High computation/communication cost; limited scalability; reward may be exploited; HE requirement limits practicality
FLBuff [28]	In-agg.	Robust Agg.	Extract penultimate layer representations (PLRs) via small auxiliary dataset; use supervised contrastive learning to build buffer layer; aggregate with MMD trust scores	Standard	Yes	Needs auxiliary dataset; supervised training; assumes trusted server; 50% malicious clients
A4FL [29]	Pre-agg.	Anomaly Det.	Adversarial training with Messy/Wrap + UAPs; neuron pruning; statistical testing on updates (variance, cosine, Euclidean)	Standard	Yes	High computational overhead
CrowdGuard [8]	Pre-agg.	Anomaly Det.	TEE-based client feedback; HLBIM validates hidden layers; iterative pruning; stacked clustering to filter poisoned updates	TEE-Based	Yes	Requires TEE hardware; client validation adds latency (29.5s/round); memory and communication overhead

Validation-Based Detection: Approaches such as [32] treat the server as an auditor, evaluating incoming updates on a small, clean validation dataset. These methods operate on the premise that malicious updates, often overfitted to backdoor triggers, exhibit higher loss or lower accuracy on benign data compared to honest clients.

Parameter Magnitude Checks: Defenses like [33] analyze the weight updates directly. Since effective poisoning often requires scaling up the update vector to overpower the global average (model replacement), these methods flag clients whose updates show excessive Euclidean distance (L_2 norm) from the global model.

Latent Space Inspection: More sophisticated methods like Fed-LSAE [34] use Autoencoders to learn the low-dimensional manifold (latent space) of benign traffic. Malicious updates containing backdoor patterns fail to compress into this learned space, resulting in high reconstruction errors that signal an anomaly.

Statistical Tail Analysis: Rather than using arbitrary thresholds, recent works [35] apply Extreme Value Theory (EVT) by fitting a Weibull distribution to validation scores. This allows the system to statistically calculate the probability that a specific client's update is an outlier relative to the historical distribution of benign updates.

Limitation: The core assumption of this category is its primary weakness. These defenses presume that benign clients are "normal" and attackers are "anomalous." This assumption breaks down in the context of a real-world NIDS, where the data is naturally non-IID (non-identically distributed). In a non-IID setting, benign clients (e.g., from an "engineering" department vs. a "marketing" department) will produce very different, "anomalous-looking" updates, leading to a high false-positive rate where good clients are mistakenly rejected.

B. Robust Aggregation-Based Defenses

Instead of acting as a "bouncer" that either accepts or rejects clients, this second category of defense acts as a "statistical gatekeeper." These methods aggregate all client updates but replace the standard averaging function with a robust alternative designed to minimize the statistical influence of outliers.

Classic approaches, reviewed in [6], generally fall into two mathematical classes:

- **Distance-Based Selection:** Algorithms like *Krum* [36] and *Multi-Krum* [36] treat the model update as a single vector. They calculate the sum of Euclidean distances between a client and its nearest neighbors, selecting

the one update (the medoid) that minimizes this spatial distance.

- **Coordinate-Wise Statistics:** Methods like *Median* [37] and *Trimmed Mean* [37] operate on each model parameter individually. For every weight in the neural network, the server sorts the values received from all clients and either selects the median value or removes the extreme tails (e.g., the top and bottom $\beta\%$) before averaging.

More recent works have proposed adaptive aggregation rules. [38] proposed FedMADE, a dynamic aggregation method that clusters clients by their traffic patterns and aggregates their models based on their contribution to overall performance. Similarly, [39] proposed a "Privacy Robust Aggregation Method" that filters outliers based on approximate Euclidean distance.

Limitation: While these methods provide a strong defense against noisy poisoning attacks (aimed at degrading accuracy), they are vulnerable to stealthy backdoor attacks. As shown in Section II, an attacker can employ "model replacement" techniques, scaling their malicious update to be mathematically "close" to the benign center in Euclidean space while still influencing the global model direction. Furthermore, coordinate-wise methods (like Median [37]) often fail when the backdoor affects a large number of parameters simultaneously, as the statistical center shifts toward the attack vector.

C. Privacy-Enhancing Defenses

The third category of defense leverages privacy-preserving techniques, not for privacy alone, but for their incidental robustness against poisoning.

The standard approach is Differential Privacy (DP). As detailed by [40], DP in Federated Learning operates through a two-step mechanism: *Gradient Clipping* and *Noise Injection*. First, the server (or client) clips the norm of the model update to a predefined threshold C , effectively bounding the maximum influence any single participant can have on the global model. Second, calibrated noise (typically Gaussian) is added to the clipped update. This process not only prevents the inference of private data but also neutralizes the "scaling" techniques used in backdoor attacks by capping the magnitude of the malicious update [41]. Recent works like [42] have further formalized this as a defense strategy.

Limitation: The primary drawback is the "utility-robustness trade-off." To theoretically guarantee protection against strong backdoors, the required noise variance often becomes so high that it degrades the model's ability to detect subtle intrusion patterns (main task accuracy). [43] investigate this tension, arguing that in high-dimensional NIDS data, the noise might hinder the defense mechanisms more than it hides the attack. Consequently, hybrid approaches like [42] attempt to mitigate this by combining mild DP with Out-of-Distribution (OOD) detection.

D. Other and Hybrid Defenses

Finally, a distinct body of research explores non-statistical architectures that rely on system design rather than data

analysis.

Ledger-Based Accountability: To address the lack of trust in clients, frameworks like FLEDGE [27] utilize blockchain technology. By recording model updates on an immutable ledger, these systems introduce accountability; while they may not prevent the initial injection of a backdoor, they ensure that malicious contributions can be traced back to specific clients, enabling reputation-based exclusion in future rounds.

Cryptographic and Hardware Security: Other approaches focus on secure computation. [44] employs cryptographic methods like Homomorphic Encryption (HE) to ensure secure aggregation, though this often complicates the server's ability to inspect individual updates for anomalies. Alternatively, CrowdGuard [8] leverages Trusted Execution Environments (TEEs) (e.g., Intel SGX) at the client side. This hardware-based approach ensures that the training code has not been tampered with, providing a "proof of training" that guarantees the client actually ran the certified algorithm on their local data.

Limitation: These architectural solutions introduce significant systemic barriers. Blockchain imposes high latency and storage overhead unsuitable for real-time NIDS, while HE requires massive computational resources. Similarly, TEE-based defenses rely on clients possessing specific hardware, limiting the scalability of the federated network in heterogeneous IoT environments.

IV. IDENTIFIED RESEARCH GAPS AND CHALLENGES

Our review of the state-of-the-art in backdoor attacks and defenses (Sections 2 and 3) reveals that while significant progress has been made, the field suffers from a critical disconnect between theoretical defense design and practical application, particularly for a domain like Network Intrusion Detection Systems (NIDS). We have identified three primary research gaps.

A. The Dataset Mismatch: Image Benchmarks vs. Network Reality

The most significant gap identified is the overwhelming reliance on image-based datasets for evaluating defense mechanisms. The vast majority of state-of-the-art defenses, including prominent works like FLAME [10] and many of its contemporaries, are designed, tested, and validated almost exclusively on simple, low-dimensional, and relatively homogeneous datasets such as MNIST, CIFAR-10, and Fashion-MNIST.

Real-world network traffic data, as required for an NIDS, bears little resemblance to these benchmarks. NIDS datasets (e.g., CIC-IDS-2017, UNSW-NB15) are high-dimensional (often with hundreds of features), tabular, highly sparse, and represent a fundamentally different data modality. There is no evidence to suggest that a defense's performance on a 32x32 pixel image dataset will translate to its ability to find a malicious backdoor hidden within a high-dimensional network flow vector. The effectiveness of these defenses in a non-image context remains almost entirely unverified.

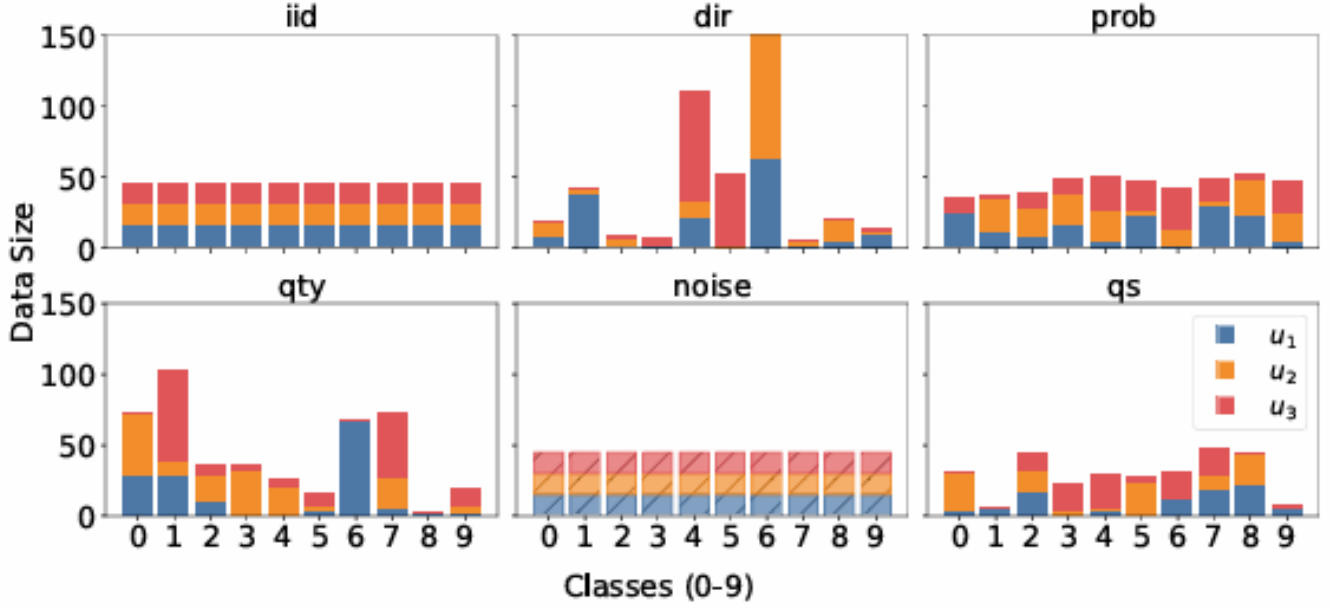


Fig. 2. Illustrations of IID and non-IID scenarios

B. The Flawed "Normality" Assumption in Non-IID Environments

The second critical gap is the "non-IID problem." Most Anomaly Detection-Based Defenses (Section 3.1) are built on a core assumption: that benign clients are statistically similar to each other, and malicious clients are "anomalous" outliers [32]. In a real-world federated NIDS, this assumption is fundamentally invalid.

Data in a large network is naturally non-identically distributed (non-IID). For example, the network traffic from an organization's "Engineering" department (using SSH, Git) will look completely different from the "Marketing" department's traffic (using web and social media). Both are benign, but in an anomaly detection model, the "Marketing" client's update would look like a severe statistical outlier compared to the "Engineering" cluster, leading to a high false-positive rate where benign clients are incorrectly filtered out.

Several non-IID scenarios remain underexplored. They are Dirichlet distribution-based label imbalance (dir), Probability-based label imbalance (prob), Quantity-based label imbalance (qty), Noise-based feature distribution (noise), Quantity skew (quantity-skew) [28].

FLAME [10] has stated it has used non-IID conditions. But has only considered Probability-based label imbalance [28], which could fail in realistic environments. Further, SHIELD-FL [11] states that it uses a non-IID setting, but does not mention which non-IID method was applied. This makes the work difficult to reproduce. Clearly specifying the non-IID scenario used—and evaluating the defense under multiple such scenarios is still limited in this research area.

C. Need for Backdoor-Specific Evaluation Metrics in FL Defense

SHIELD-FL [11] uses a client-side defense through Neuron Synaptic Weight Adjustment. Although they define backdoor specific metric to find unstable neurons, They only evaluate there defense on basic metrics such as Accuracy, precision, recall, F1-score which has no correlation with backdoor specific metrics.

FLAME [10] has evaluated its defense with Backdoor Accuracy (BA). it clearly shows without defense, they have 100% Main Task Accuracy and also 100% Backdoor Accuracy. This shows improving Accuracy won't remove Backdoors. There is a crucial limitation on evaluating the backdoor defense frameworks with backdoor specific metrics such as ASR(attack success rate), BA(backdoor accuracy).

D. Security and Privacy Risks of Auxiliary Datasets in FL Defense

SHIELD-FL [11] defense needs an auxiliary dataset in each client. This is acceptable from a privacy perspective but it introduces a major security issue because malicious clients can poison their auxiliary datasets. FLBuff [28] has introduced a method which is doing the defense in server side. But they also need an auxiliary dataset. which violates the privacy of users. Making the defense framework without the auxiliary dataset is also crucial.

E. The Application-Defense Disconnect

Finally, we observe a disconnect from the other direction. The body of literature that does apply Federated Learning to NIDS (as seen in Section 2, e.g., [3]; [4]) is primarily focused on solving the problems of privacy and non-IID

accuracy. These papers are rightly concerned with proving that a federated NIDS can be as accurate as a centralized one.

However, these application-focused papers rarely, if ever, rigorously evaluate their proposed systems against the sophisticated, stealthy backdoor attacks detailed in Section 2. Their threat models are often limited to simple poisoning or no attacks at all. As a result, the field has (1) a set of strong backdoor defenses that haven't been tested on NIDS data, and (2) a set of FL-NIDS systems that haven't been tested against strong backdoor attacks.

V. CONCLUSION AND FUTURE WORK

Federated Learning (FL) presents a powerful solution to the critical privacy challenges inherent in traditional, centralized Network Intrusion Detection Systems (NIDS). By enabling collaborative model training without sharing sensitive network logs, FL offers a path toward building robust detection models that respect user and organizational privacy. However, as this review has demonstrated, the distributed nature of FL introduces a severe vulnerability to sophisticated, stealthy backdoor attacks.

We have categorized and analyzed the state-of-the-art defenses, which primarily fall into three categories: anomaly detection, robust aggregation, and differential privacy. Our analysis reveals a critical disconnect: the majority of existing research fails to validate defenses under multiple non-IID (non-identically distributed) scenarios. Since backdoor attacks specifically exploit the non-IID nature of local data, it is imperative that future defenses are proven effective under diverse and realistic data distribution settings.

we propose the following directions for future research:

A. NIDS-Centric Defense Validation: Future research must move beyond standard image-based benchmarks (e.g., MNIST, CIFAR-10) and validate defenses on high-dimensional, sparse, and tabular network traffic datasets (e.g., CIC-IDS-2017, UNSW-NB15). Network traffic features exhibit different statistical correlations than pixel data. Therefore, defense mechanisms must be tailored to detect trigger patterns specifically within packet flow features rather than spatial visual patterns.

B. Robustness in Non-IID Environments: The assumption that benign clients are statistically similar is fundamentally invalid in large-scale networks where traffic patterns vary significantly by department or device type. Future work should prioritize distribution-agnostic defenses. Instead of flagging diverse clients as anomalies, researchers should explore techniques such as personalized federated learning or meta-learning that can distinguish between benign distributional shifts (concept drift) and malicious model updates.

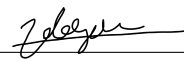
C. Standardization of Backdoor Metrics: Evaluating systems solely on main-task accuracy is insufficient for security-critical applications. A defense that maintains high accuracy but allows a backdoor to persist is a failure. The community should adopt a standardized evaluation framework that mandates the reporting of Attack Success Rate (ASR) and Backdoor Accuracy (BA) alongside standard metrics like F1-score to validate true resilience against targeted attacks.

D. Zero-Trust Defense Architectures: Current defenses often rely on "clean" auxiliary datasets or trusted validation data on the client or server side, which introduces a new attack surface. In a zero-trust environment, malicious clients can poison these auxiliary sets. Future frameworks should eliminate the need for trusted data, potentially by exploring unsupervised server-side auditing, cryptographic proofs of update integrity, or robust aggregation methods that operate purely on the statistical properties of the updates themselves.

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